



Zurich
Instruments

MFLI
Lock-in Amplifier
500 kHz / 5 MHz

Introducing the MFLI

What is it and who is it for?

- Background (need to add pot. Markets?)
- Product View
- Key Specifications
- LabOne®

Product Comparison

- Specification comparison with competitors

Pricing

Availability

Background

Why the MFLI?

- Our customers have been asking for an instrument which is targeted at a lower frequency range than our HF2LI and UHFLI products, and with only a single lock-in unit, but don't want to compromise on performance or features.
- It enables us to compete at the lower price end of the lock-in market, whilst retaining ZI's premium performance, unique features and functionality
- It gives ZI a complete lock-in product offer, with no price point or specification level left untouched
- In turn, it empowers our sales partners with the knowledge that they have a product suitable for every lock-in customer
- It incorporates all of our knowledge up to this point, making it *the* state of the art lock-in amplifier

MFLI – Front Panel



Features:

- Plastic “bumper” for easy handling and stacking
- 2U height, half the width of HF2LI/HF2IS/UHFLI instruments
- Rack mount available; “side by side” layout possible
- Prepared for integration business

MFLI – Rear Panel



Features:

- ─ Note USB storage capability
- ─ DC capability – OEM potential. Can use battery power too.
- ─ Clock in/out allows synchronization with other instruments or clocks
- ─ IEEE1588 synchronization possible – do customers ask for this?

Key Specifications

Frequency Range

- DC to 5 kHz (MFLI 500k) or
- DC to 5 MHz (MFLI 5M); MFLI 500k upgradable to 5 MHz with the MF-F5M option

Time constant/measurement bandwidth

- 500 ns to 76s, 80 μ Hz to 200kHz

Best-in-class Performance

- 3 nV/ $\sqrt{\text{Hz}}$ input noise

Inputs/Outputs

- 1x voltage input (single ended or differential), 1x current input
- 1x voltage output (single ended or differential)

Connectivity & Software

- MFLI comes with LabOne® and its toolset: see next slide
- USB 2.0 and 1 GbE high speed connections

LabOne®

MFLI comes with LabOne®

- Browser based UI
 - The same UI as the UHFLI
 - No additional PC required
 - This is new: all that's required is a network connection
- Toolset
 - Spectrum analyzer, scope, sweeper, plotter, software trigger etc.
- Connectivity
 - USB 2.0 (available Q3), 1GbE
 - Remote operation possible (PC, tablet, phone)
- Programmability
 - APIs for LabView, MATLAB, C and Python

USPs

- Industry leading specifications, functionality and frequency range at this price point
 - Unbeatable “complete” package
- Usability
 - Complete measurement toolbox with LabOne[®], saving on instrument costs
- Connectivity
 - Easy data transfer and simple “Plug and Play” setup. Instant measurement!
- Upgradeability
 - Options can be added in the field – the instrument “grows” with you
- Programmability
 - Same APIs across the product range,
- Portability
 - Platform independent for remote operation

Competitor Comparison

	Stanford SR830	Stanford SR865 (in 2015?)	Signal Recovery SR7280	Anfatec eLockIn 204	Zurich Instruments MFLI
Input frequency Range	1 mHz – 102.4 kHz	1 mHz – 2 MHz	0.5Hz – 2 MHz	1 Hz – 1 MHz	DC – 5 MHz ^(a)
Dynamic Reserve	>100dB	N/A	>100dB	>135dB ^(b)	120dB
Minimum Time Constant	10 μ s	N/A	50 μ s	100 μ s	500 ns
Demodulators	1	1	1	1	Up to 4
Connectivity	GPIO, RS-232	GPIO, RS-232, Ethernet, USB	GPIO, RS-232	Ethernet, USB	USB 2.0 (from Q3), 1GbE
Software inclusive?	No	No	No	Yes	LabOne®
Additional instrumentation	No	2	No	2	> 4 (including spectrum analyzer)
Options available?	No	No	No	No	Yes
Digital Output to PC	0.512 kHz	N/A	1 kHz	Unknown?	200 kHz

(a) MFLI is available with two frequency ranges, from DC to 500 kHz or DC to 5 MHz. (b) Anfatec have an odd way of specifying dynamic reserve which is not universally accepted – this figure is unlikely to be “true”. (c) This prices depends on the variant chosen. (d) This price is the USD equivalent of the EUR price

Zurich Instruments

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